

8 numbers problem

Tags:

Time Limit: 1 s Memory Limit: 64 MB
Submission: 350 AC: 96 Score: 96.61

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AI Code Tips

Description

I think almost every acmer will know the 8 numbers problem which is a very famous problem. The game begin from the initial state of a 3*3 matrix which makes up of 8 numbers(1-8) and a blank block(0). Move the blank block with its adjacent block until reach the objective state. It is obvious that the blank block has four directions which it can move to when it is at the middle position, i.e. up, down, left, right. Also, it has two directions when it is at the corner of the matrix and three directions at other position.

For example,the initial state of the matrix:

```
8 0 3
2 1 4
7 6 5
```

the objective state:

```
1 2 3
8 0 4
7 6 5
```

and we give a valid moving path:

```
8 0 3      8 1 3      8 1 3      0 1 3      1 0 3      1 2 3
2 1 4 > 2 0 4 > 0 2 4 > 8 2 4 > 8 2 4 > 8 0 4
7 6 5      7 6 5      7 6 5      7 6 5      7 6 5      7 6 5
```

Moreover,the path with least steps is called the shortest path. And the 8 numbers is check whether there are the path from the initial state to the objective state and if it exists, give the shortest path.

And we all know huicpc229 is not very good at search, so he hasn't solved this problem now. But he has solved another easy problem. The problem is described as follow:

Give an initial state of the matrix, and give a sequence of moving. For every moving, if the blank block can move to the direction as the moving, move it, otherwise ignore this moving. And we want to know the final state of the matrix.

Input

The first line of the input is one integer t, the number of test cases.

For each test case:

Three lines respond to the initial state of the matrix, and there will be three numbers on each of the three lines.

Next line follow by an integer m corresponding to the number of moving.

The next m line, every line contain only one character:

U: move the blank block up for one block.

D: move the blank block down for one block.

L: move the blank block left for one block.

R: move the blank block right for one block.

Output

For each test case output the final state of the matrix for three lines as above. And there will be a blank space between every two numbers on the same line. And you should output one blank line after each test case.

Samples

input	Copy
<pre>1 8 0 3 2 1 4 7 6 5 2 D R</pre>	
output	Copy
<pre>8 1 3 2 4 0 7 6 5</pre>	

Source

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